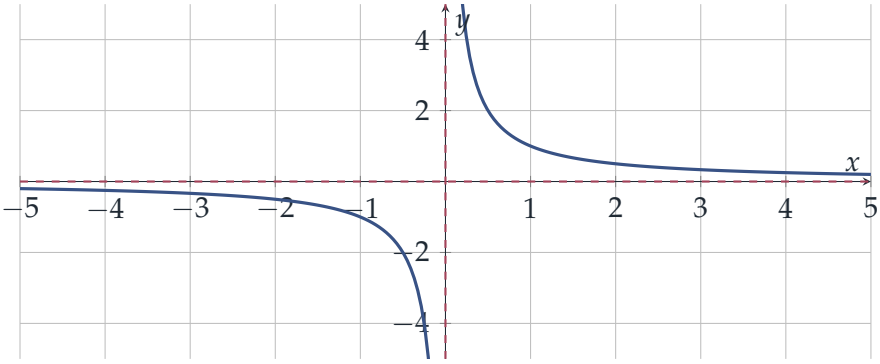
Chapter 4

Reciprocal function

Skills: determine a domain, differentiate, study a sign, draw up a table of variations, represent

4.1 Reference notes

The reciprocal function is defined by $f(x) = \frac{1}{x}$ on $\mathbb{R} \setminus \{0\}$.



Key ideas

$f'(x) = -\frac{1}{x^2} < 0$ for every $x \neq 0$. The reciprocal function is strictly decreasing on each of the intervals $]-\infty, 0[$ and $]0, +\infty[$. The lines $x = 0$ and $y = 0$ are asymptotes.

Common mistake

We cannot write that the reciprocal function is decreasing "on $\mathbb{R} \setminus \{0\}$" without specifying the two intervals. For example $-1 < 1$ but $f(-1) = -1 < 1 = f(1)$, which would contradict the definition of a decreasing function on the whole set.

For a function $g(x) = P(x) + \frac{k}{x}$, proceed in this order: domain; derivative; common denominator and factorisation; sign of the derivative; variations; limits or behaviour at the bounds; useful points; representation.

Worked example

On $]0, +\infty[$, $g(x) = x + \frac{4}{x}$ satisfies $g'(x) = 1 - \frac{4}{x^2} = \frac{(x-2)(x+2)}{x^2}$. The denominator is positive; g' is negative on $]0, 2[$ then positive on $]2, +\infty[$. The function has a minimum $g(2) = 4$.

Session 23 — Definition and graph of the reciprocal function

Reciprocal function

15/12/2026 • B : 2 h – C : 1 h

Objectives of the session

· Determine the domain.

· Draw the curve.

· Use symmetries.

Fluency · 6 to 8 minutes, without a calculator

1. Simplify 3 $x + 5 - x + 2$. $2x + 7$

2. Give the forbidden value of $\dfrac{x+1}{x}$. $x = 0$

3. Calculate $\dfrac{1}{5}$ and ~~-1~~ $\dfrac{1}{5}$. $\dfrac{1}{5}$ and $-\dfrac{1}{5}$

4. Differentiate 3 $x^2 - 5x + 1$. $6x - 5$

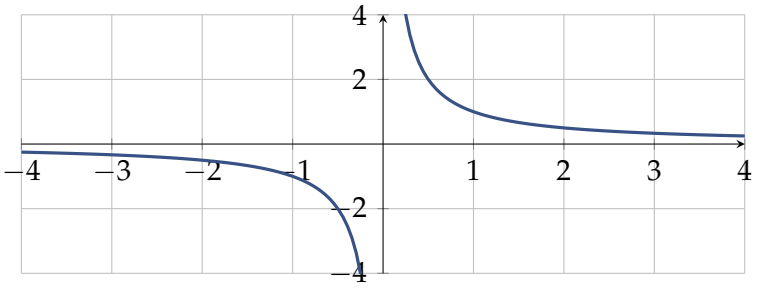
5. Solve $(x-2)(x+1) = 0$. $x = -1$ **or** $x = 2$

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: determine the domain and draw the graph.

Investigation

Calculate $\dfrac{1}{x}$ for $x = -4, -2, -1, -\dfrac{1}{2}, \dfrac{1}{2}$, 1, 2, 4. Plot the points and describe the symmetries. Why $x = 0$ missing?



Detailed solutions · teacher edition

The values are respectively $-\frac{1}{4}, -\frac{1}{2}, -1, -2, 2, 1, \frac{1}{2}, \frac{1}{4}$. The domain excludes 0; the curve is symmetric about the origin. Have students put into words the information that triggers the choice of method.

Written summary

The reciprocal function is $f(x) = \dfrac{1}{x}$ on $\mathbb{R} \setminus \{0\}$. Its graph is a hyperbola symmetric about the origin.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Detailed solutions · teacher edition

The values are respectively $-\frac{1}{4}, -\frac{1}{2}, -1, -2, 2, 1, \frac{1}{2}, \frac{1}{4}$. The domain excludes 0; the curve is symmetric about the origin.

Practice · essentials

1. Give the domain and the asymptotes of $f(x) = 5 + \dfrac{7}{x}$.
2. Calculate $f'(x)$ and study its sign.
3. Study on $]0, +\infty[$ the function $g(x) = x + \dfrac{25}{x}$.

Detailed solutions · teacher edition

The domain is $\mathbb{R} \setminus \{0\}$; asymptotes $x = 0$ and $y = 5$. $f'(x) = -\dfrac{7}{x^2} < 0$. For g, $g'(x) = 1 - \dfrac{25}{x^2}$ and the minimum is attained at $x = 5$, with value 10.

References in the textbook

Extensions: Calao, lesson pp. 18-21; exercises p. 24, nos. 1-3. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find the figures and the textbook's digital resources.

Calao textbook · p. 101, Exercise 42

The aim of this exercise is to draw the graphs of the functions f and g defined on $\mathbb{R}^*$ by: $f(x) = \dfrac{2}{x}$ and $g(x) = 4 - \dfrac{3}{x}$.

1. Complete the table of values for $x \in \{-5; -4; -2; -0{,}5; -0{,}2; -0{,}1; 0{,}1; 0{,}2; 0{,}5; 1; 2; 4; 5\}$.
2. Using a table of values, draw the graphs of the functions f and g.

Detailed solutions · teacher edition

Calculate each image, respecting the forbidden value 0. The two curves have the vertical asymptote $x = 0$. The horizontal asymptote of f is $y = 0$; that of g is $y = 4$. To check the drawing, check for example $f(1) = 2$, $f(-1) = -2$, $g(1) = 1$ and $g(-1) = 7$.

Independent work

Redo practice exercise 2 with the following data: replace the initial value with 235 and the rate or common ratio with 7. Write four lines: data, formula, calculation, conclusion.

Detailed solutions · teacher edition

The solution depends on the exercise revisited. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Method

Lesson plan for 1 h: 7 min of fluency; 10 min of sharing and written summary; 28 min of core exercises; 5 min of correction; 3 min for the exit ticket.

Lesson plan for 2 h: add 20 min of activity work, 20 min of formalising the lesson, a second independent exercise and 10 min of comparative correction.

Differentiated teaching: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast students to produce a counterexample, a justification or a second method.

Session 24 — Behaviour at the endpoints and asymptotes

Reciprocal function

17/12/2026 • B : 1 h – C : 2 h

Objectives of the session

· Describe the behaviour at the endpoints.

· Identify the asymptotes.

· Read a representation.

Fluency · 6 to 8 minutes, without calculator

1. Calculate 5% of 120. 6

2. Calculate $\frac{1}{6}$ and ~~$-\frac{1}{6}$~~. $\frac{1}{6}$ and $-\frac{1}{6}$

3. Differentiate $x^2 - 5x + 1$. $6x - 5$

4. Solve $(x-2)(x+1) = 0$. $x = -1$ **or** $x = 2$

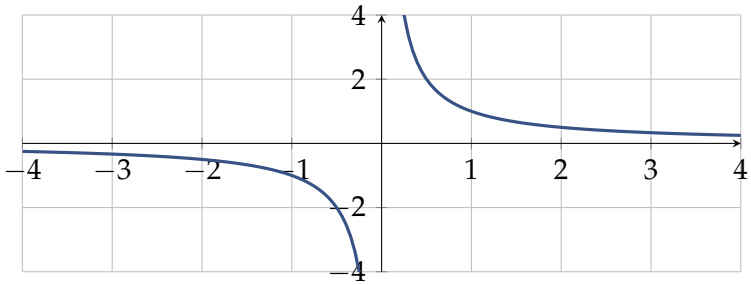
5. Give the forbidden value of $\frac{x+1}{x}$. $x = 0$

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: describe the behaviour at the endpoints and identify the asymptotes.

Investigation

Using a table of values close to 0 and for large values of $|x|$, describe $\frac{1}{x}$. Express each behaviour with an asymptote and a sentence.



Detailed solutions · teacher edition

Near 0, the values become very large in absolute value: asymptote $x = 0$. When $|x|$ grows, $\frac{1}{x}$ approaches 0: asymptote $y = 0$. Have students verbalise the information that triggers the choice of method.

Written summary

For $f(x) = \frac{1}{x}$, $x = 0$ is a vertical asymptote and $y = 0$ is a horizontal asymptote. The branches

approach them without crossing them.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Detailed solutions · teacher edition

Near 0, the values become very large in absolute value: asymptote $x = 0$. When $|x|$ grows, $\dfrac{1}{x}$ approaches 0: asymptote $y = 0$.

Practice · essentials

1. Give the domain and the asymptotes of $f(x) = 6 + \dfrac{8}{x}$.
2. Calculate $f'(x)$ and study its sign.
3. Study on $]0, +\infty[$ the function $g(x) = x + \dfrac{36}{x}$.

Detailed solutions · teacher edition

The domain is $\mathbb{R} \setminus \{0\}$; asymptotes $x = 0$ and $y = 6$. $f'(x) = -\dfrac{8}{x^2} < 0$. For g, $g'(x) = 1 - \dfrac{36}{x^2}$ and the minimum is attained at $x = 6$, with value 12.

References in the textbook

Extensions: Calao, lesson pp. 18-21; exercises p. 24, nos. 1-3. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find the figures and the textbook's digital resources.

Calao textbook · p. 95, Exercise 16

We want to plot the graph of the function g defined on $\mathbb{R}^*$ by: $g(x) = 3 - \dfrac{2}{x}$.

1. Complete a table of values for

$$x \in \{-1\,000; -500; -200; -100; -50; 50; 100; 200; 500; 1\,000\}.$$

   What can we deduce about the behaviour of g at $-\infty$ and at $+\infty$?

2. Complete a table of values for $x = -0{,}5; -0{,}1; -0{,}01; 0{,}01; 0{,}1; 0{,}5$. What can we deduce about the behaviour of g at 0?
3. Copy and complete the table of values from the textbook, then plot the curve of the function g.

Detailed solutions · teacher edition

When $x \to \pm\infty$, $-2/x \to 0$, therefore $g(x) \to 3$. When $x \to 0^-$, $-2/x \to +\infty$; when $x \to 0^+$, $-2/x \to -\infty$. The asymptotes are $x = 0$ and $y = 3$.

Independent work

Redo practice exercise 3 with the following data: replace the initial value with 240 and the rate or common ratio with 2. Write four lines: data, formula, calculation, conclusion.

Detailed solutions · teacher edition

The solution depends on the exercise revisited. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Method

Lesson plan for 1 h: 7 min of fluency; 10 min of sharing and written summary; 28 min of core exercises; 5 min of correction; 3 min for the exit ticket.
Lesson plan for 2 h: add 20 min of activity work, 20 min of formalising the lesson, a second independent exercise and 10 min of comparative correction.
Differentiated teaching: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast students to produce a counterexample, a justification or a second method.

Session 25 — Derivative and direction of variation

Reciprocal function

05/01/2027 • B : 2 h – C : 1 h

Objectives of the session

· Differentiate the reciprocal function.

· Justify the variations.

· Draw up a table.

Fluency · 6 to 8 minutes, without calculator

1. Write the multiplier for a 10% increase. $1+\frac{10}{100}=1,10$

2. Differentiate 3 x^2-5x+1. $6x-5$

3. Solve $(x-2)(x+1)=0$. $x=-1$ **or** $x=2$

4. Give the forbidden value of $\frac{x+1}{x}$. $x=0$

5. Calculate $\frac{1}{7}$ and $-\frac{1}{7}$. $\frac{1}{7}$ and $-\frac{1}{7}$

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: differentiate the reciprocal function and justify the variations.

Investigation

From the rate of change of $f(x)=\frac{1}{x}$, deduce $f'(x)=-\frac{1}{x^2}$. Study its sign and check consistency with the graph.

Detailed solutions · teacher edition

The square x^2 is positive for $x\neq 0$, so the derivative is strictly negative on each of the intervals of the domain. Have students verbalise the information that triggers the choice of method.

Written summary

For $x\neq 0$, $f'(x)=-\frac{1}{x^2}<0$. The reciprocal function decreases on each of the intervals $]-\infty,0[$ and $]0,+\infty[$.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Detailed solutions · teacher edition

The square x^2 is positive for $x \neq 0$, so the derivative is strictly negative on each of the intervals of the domain.

Practice · essentials

1. Give the domain and the asymptotes of $f(x) = 2 + \dfrac{4}{x}$.
2. Calculate $f'(x)$ and study its sign.
3. Study on $]0, +\infty[$ the function $g(x) = x + \dfrac{4}{x}$.

Detailed solutions · teacher edition

The domain is $\mathbb{R} \setminus \{0\}$; asymptotes $x = 0$ and $y = 2$. $f'(x) = -\dfrac{4}{x^2} < 0$. For g, $g'(x) = 1 - \dfrac{4}{x^2}$ and the minimum is attained at $x = 2$, with value 4.

References in the textbook

Extensions: Calao, lesson pp. 20-21; exercises p. 24, nos. 11-13, then p. 28, nos. 43-44. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find the figures and the textbook's digital resources.

Calao textbook · p. 94, Exercise 3

Calculate the derivatives of the following functions defined on $\mathbb{R}^*$.

a. $f(x) = 2x - 3 + \dfrac{4}{x}$

b. $g(x) = 3x - 5 - \dfrac{1}{x}$

c. $f(x) = 7x^3 - 2x + \dfrac{1}{x}$

d. $g(x) = 3x^2 - 5x - 2 + \dfrac{4}{x}$

Detailed solutions · teacher edition

The derivatives are: $f'(x) = 2 - 4/x^2$; $g'(x) = 3 + 1/x^2$; $f'(x) = 21x^2 - 2 - 1/x^2$; $g'(x) = 6x - 5 - 4/x^2$.

Independent work

Redo practice exercise 2 with the following data: replace the initial value with 245 and the rate or common ratio with 3. Write four lines: data, formula, calculation, conclusion.

Detailed solutions · teacher edition

The solution depends on the exercise revisited. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Method

Lesson plan for 1 h: 7 min of fluency; 10 min of sharing and written summary; 28 min of core exercises; 5 min of correction; 3 min for the exit ticket.

Lesson plan for 2 h: add 20 min of activity work, 20 min of formalising the lesson, a second independent exercise and 10 min of comparative correction.

Differentiated teaching: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast students to produce a counterexample, a justification or a second method.

Session 26 — Combinations with polynomial functions

Reciprocal function

07/01/2027 • B : 1 h – C : 2 h

Objectives of the session

· Differentiate a combination.

· Simplify an expression.

· Organise a study.

Fluency · 6 to 8 minutes, without a calculator

1. Simplify 3 $x + 8 - x + 2$. $2x + 10$

2. Solve $(x - 2)(x + 1) = 0$. $x = -1$ **or** $x = 2$

3. Give the forbidden value of $\dfrac{x+1}{x}$. $x = 0$

4. Calculate $\dfrac{1}{8}$ and ~~$\dfrac{1}{8}$~~. $\dfrac{1}{8}$ and $-\dfrac{1}{8}$

5. Differentiate 3 $x^2 - 5x + 1$. $6x - 5$

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: differentiate a combination and simplify an expression.

Investigation

A company produces $x > 0$ hundred objects and its average cost is $C(x) = 2x + 30 + \dfrac{200}{x}$. Calculate $C'(x)$, factorise its numerator and find the production level that minimises the average cost.

Detailed solutions · teacher edition

$C'(x) = 2 - \dfrac{200}{x^2} = \dfrac{2(x-10)(x+10)}{x^2}$. The minimum is attained at $x = 10$, i.e. 1,000 objects. Have students put into words the information that triggers the choice of method.

Written summary

To study $P(x) + \dfrac{k}{x}$, determine the domain, calculate the derivative, reduce to the same denominator, study its sign then draw up the variations.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Detailed solutions · teacher edition

$C'(x) = 2 - \dfrac{200}{x^2} = \dfrac{2(x-10)(x+10)}{x^2}$. The minimum is attained at $x = 10$, i.e. 1,000 objects.

Practice · essentials

1. Give the domain and the asymptotes of $f(x) = 3 + \dfrac{5}{x}$.
2. Calculate $f'(x)$ and study its sign.
3. Study on $]0, +\infty[$ the function $g(x) = x + \dfrac{9}{x}$.

Detailed solutions · teacher edition

The domain is $\mathbb{R} \setminus \{0\}$; asymptotes $x = 0$ and $y = 3$. $f'(x) = -\dfrac{5}{x^2} < 0$. For g, $g'(x) = 1 - \dfrac{9}{x^2}$ and the minimum is attained at $x = 3$, with value 6.

References in the textbook

Extensions: Calao, p. 96, No. 14-21. The exercise reproduced in this session is identified below. by its page and number; the reference also lets you find the figures and the digital resources of the textbook.

Calao textbook · p. 96, Exercise 20

Let f the function defined on $\mathbb{R}^*$ by $f(x) = 4x + 1 + \dfrac{9}{x}$.

1. Show that for every real x non-zero: $f'(x) = \dfrac{(2x-3)(2x+3)}{x^2}$.
2. For every real x non-zero, study the sign of $f'(x)$.
3. Deduce the variation of the function f.

Detailed solutions · teacher edition

$f'(x) = 4 - 9/x^2 = (4x^2 - 9)/x^2 = ((2x-3)(2x+3))/x^2$. The denominator is positive: f' is positive on $]-\infty; -3/2[\cup]3/2; +\infty[$ and negative on $]-3/2; 0[\cup]0; 3/2[$. We deduce the variations on each of the intervals separated by $-3/2$, 0 and 3/2.

Independent work

Redo practice exercise 3 with the following data: replace the initial value with 250 and the rate or common ratio with 4. Write four lines: data, formula, calculation, conclusion.

Detailed solutions · teacher edition

The solution depends on the exercise revisited. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Method

Lesson plan for 1 h: 7 min of fluency; 10 min of sharing and written summary; 28 min of core exercises; 5 min of correction; 3 min for the exit ticket.

Lesson plan for 2 h: add 20 min of activity work, 20 min of formalising the lesson, a second independent exercise and 10 min of comparative correction.

Differentiated teaching: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast students to produce a counterexample, a justification or a second method.

Session 27 — Sign and factorisation

Reciprocal function

12/01/2027 • B : 2 h – C : 1 h

Objectives of the session

· Factorise.

· Build a sign table.

· Respect the forbidden values.

Fluency · 6 to 8 minutes, without calculator

1. Calculate 20% of 180. 36
2. Give the forbidden value of $\dfrac{x+1}{x}$. $x = 0$
3. Calculate $\dfrac{1}{9}$ and ~~$-\dfrac{1}{9}$~~. $\dfrac{1}{9}$ and $-\dfrac{1}{9}$
4. Differentiate 3 $x^2 - 5x + 1$. $6x - 5$
5. Solve $(x-2)(x+1) = 0$. $x = -1$ **or** $x = 2$

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today. Before the activity, have students rephrase the key words: factorise and build a sign table.

Investigation

Study the sign of $\dfrac{(x-2)(x+3)}{x}$. Construct the table with the three critical values, distinguish zero and forbidden value, then solve the inequality $\dfrac{(x-2)(x+3)}{x} \geq 0$.

Detailed solutions · teacher edition

The critical values are -3, 0, 2. The solution is $[-3, 0[\cup [2, +\infty[$. Have students verbalise the information that triggers the choice of method.

Written summary

In a quotient, the sign depends on the numerator and the denominator. The forbidden values are excluded from the table, even if a factor formally cancels.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Detailed solutions · teacher edition

The critical values are -3, 0, 2. The solution is $[-3, 0[\cup [2, +\infty[$.

Practice · essentials

1. Give the domain and the asymptotes of $f(x) = 4 + \dfrac{6}{x}$.
2. Calculate $f'(x)$ and study its sign.
3. Study on $]0, +\infty[$ the function $g(x) = x + \dfrac{16}{x}$.

Detailed solutions · teacher edition

The domain is $\mathbb{R} \setminus \{0\}$; asymptotes $x = 0$ and $y = 4$. $f'(x) = -\dfrac{6}{x^2} < 0$. For g, $g'(x) = 1 - \dfrac{16}{x^2}$ and the minimum is attained at $x = 4$, with value 8.

References in the textbook

Extensions: Calao, pp. 96-101. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and the digital resources of the textbook.

Calao textbook · p. 96, Exercise 21

Let g the function defined on $\mathbb{R}^*$ by $g(x) = x + 10 - \dfrac{1}{x}$.

1. Show that for every real x non-zero: $g'(x) = \dfrac{x^2 + 1}{x^2}$.
2. For every real x non-zero, study the sign of $g'(x)$.
3. Deduce the variation of the function g.

Detailed solutions · teacher edition

$g'(x) = 1 + 1/x^2 = (x^2 + 1)/x^2 > 0$ for every $x \neq 0$. The function is therefore strictly increasing on each of the intervals $]-\infty; 0[$ and $]0; +\infty[$.

Independent work

Redo practice exercise 2 with the following data: replace the initial value with 255 and the rate or common ratio with 5. Write four lines: data, formula, calculation, conclusion.

Detailed solutions · teacher edition

The solution depends on the exercise revisited. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Method

Lesson plan for 1 h: 7 min of fluency; 10 min of sharing and written summary; 28 min of core exercises; 5 min of correction; 3 min for the exit ticket.

Lesson plan for 2 h: add 20 min of activity work, 20 min of formalising the lesson, a second independent exercise and 10 min of comparative correction.
Differentiated teaching: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast students to produce a counterexample, a justification or a second method.

Session 28 — Derivatives of polynomials of degree at most 3

Reciprocal function

14/01/2027 • B : 1 h – C : 2 h

Objectives of the session

· Differentiate a polynomial.

· Study a derivative.

· Deduce variations.

Fluency · 6 to 8 minutes, without calculator

1. Write the multiplier for a 25% increase. $1+\frac{25}{100}=1,25$

2. Calculate $\frac{1}{3}$ and ~~$-\frac{1}{3}$~~. $\frac{1}{3}$ and $-\frac{1}{3}$

3. Differentiate x^2-5x+1. $6x-5$

4. Solve $(x-2)(x+1)=0$. $x=-1$ **or** $x=2$

5. Give the forbidden value of $\frac{x+1}{x}$. $x=0$

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: differentiate a polynomial and study a derivative.

Investigation

For $f(x)=2x^3-3x^2-12x+5$, calculate $f'(x)$, factorise, study its sign and deduce the variations of f.

Detailed solutions · teacher edition

$f'(x)=6x^2-6x-12=6(x-2)(x+1)$. The function increases on $]-\infty,-1]$, decreases on $[-1,2]$, then increases. Have students verbalise the information that triggers the choice of method.

Written summary

$(ax^3+bx^2+cx+d)'=3ax^2+2bx+c$. The sign of the derivative gives the direction of variation of the function.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Detailed solutions · teacher edition

$f'(x)=6x^2-6x-12=6(x-2)(x+1)$. The function increases on $]-\infty,-1]$, decreases on $[-1,2]$, then increases.

Practice · essentials

1. Give the domain and the asymptotes of $f(x) = 5 + \frac{7}{x}$.
2. Calculate $f'(x)$ and study its sign.
3. Study on $]0, +\infty[$ the function $g(x) = x + \frac{25}{x}$.

Detailed solutions · teacher edition

The domain is $\mathbb{R} \setminus \{0\}$; asymptotes $x = 0$ and $y = 5$. $f'(x) = -\frac{7}{x^2} < 0$. For g, $g'(x) = 1 - \frac{25}{x^2}$ and the minimum is attained at $x = 5$, with value 10.

References in the textbook

Extensions: Calao, pp. 94-96. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and the digital resources of the textbook.

Calao textbook · p. 94, Exercise 13

Let f the function defined on $\mathbb{R}^*$ by: $f(x) = \frac{1}{x}$.

1. Calculate $f'(x)$.
2. Show that for every real x non-zero: $f'(x) = \frac{-1}{x^2}$.
3. After studying the sign of $f'(x)$, draw the variation table of f.

Detailed solutions · teacher edition

$f'(x) = -1/x^2 < 0$ for every $x \neq 0$. The reciprocal function is therefore strictly decreasing on each of the intervals $]-\infty; 0[$ and $]0; +\infty[$. These two intervals should not be combined into one interval on which it decreases.

Independent work

Redo practice exercise 3 with the following data: replace the initial value with 260 and the rate or common ratio with 6. Write four lines: data, formula, calculation, conclusion.

Detailed solutions · teacher edition

The solution depends on the exercise revisited. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Method

Lesson plan for 1 h: 7 min of fluency; 10 min of sharing and written summary; 28 min of core exercises; 5 min of correction; 3 min for the exit ticket.

Lesson plan for 2 h: add 20 min of activity work, 20 min of formalising the lesson, a second independent exercise and 10 min of comparative correction.
Differentiated teaching: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast students to produce a counterexample, a justification or a second method.

Session 29 — Complete study and graph

Reciprocal function

19/01/2027 • B : 2 h – C : 1 h

Objectives of the session

· Carry out a complete study.

· Draw a coherent curve.

· Interpret an extremum.

Fluency · 6 to 8 minutes, without a calculator

1. Simplify 3 $x+4-x+2$. $2x+6$

2. Differentiate 3 x^2-5x+1. $6x-5$

3. Solve $(x-2)(x+1)=0$. $x=-1$ **or** $x=2$

4. Give the forbidden value of $\dfrac{x+1}{x}$. $x=0$

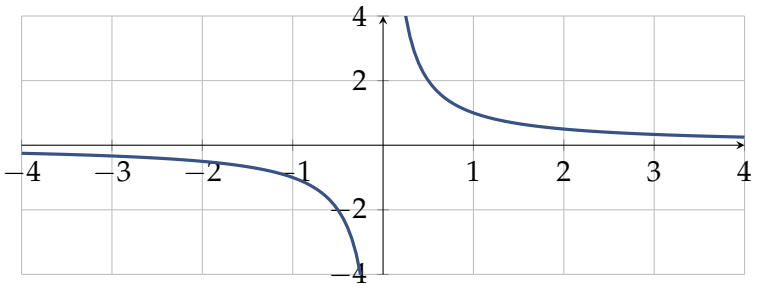
5. Calculate $\dfrac{1}{4}$ and ~~$-\dfrac{1}{4}$~~. $\dfrac{1}{4}$ and $-\dfrac{1}{4}$

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today. Before the activity, have students rephrase the key words: carry out a complete study and draw a consistent curve.

Investigation

Study fully $g(x)=x+\dfrac{9}{x}$ on $]0,+\infty[$: domain, derivative, sign, variations, minimum, behaviour at the bounds and sketch.



Detailed solutions · teacher edition

$g'(x)=\dfrac{(x-3)(x+3)}{x^2}$; minimum $g(3)=6$; $g(x)$ tends to $+\infty$ at 0^+ and at $+\infty$. Have students verbalise the piece of information that triggers the choice of method.

Written summary

A full study includes: domain, derivative, sign, variations, limits or behaviour at the bounds, notable points and a consistent graph.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Detailed solutions · teacher edition

$g'(x) = \dfrac{(x-3)(x+3)}{x^2}$; minimum $g(3) = 6$; $g(x)$ tends to $+\infty$ at 0^+ and at $+\infty$.

Practice · essentials

1. Give the domain and the asymptotes of $f(x) = 6 + \dfrac{8}{x}$.
2. Calculate $f'(x)$ and study its sign.
3. Study on $]0, +\infty[$ the function $g(x) = x + \dfrac{36}{x}$.

Detailed solutions · teacher edition

The domain is $\mathbb{R} \setminus \{0\}$; asymptotes $x = 0$ and $y = 6$. $f'(x) = -\dfrac{8}{x^2} < 0$. For g, $g'(x) = 1 - \dfrac{36}{x^2}$ and the minimum is attained at $x = 6$, with value 12.

References in the textbook

Extensions: Calao, pp. 98-103. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and the digital resources of the textbook.

Calao textbook · p. 98, Exercise 31

A company specialising in industrial ovens can make between 20 and 80 of them depending on the years. Let x the number of ovens manufactured annually.
The graph given in the textbook is that of a function C defined on the interval $[20; 80]$. For all integer values of x, $C(x)$ is the average unit production cost (in €).

1. What is the average unit production cost when 25 ovens are produced, then when 70 ovens are produced?
2. Which production levels correspond to an average unit cost of 20,000 € ?
3. What is the minimum average unit production cost? What output level does it correspond to?

Detailed solutions · teacher edition

The numerical answers are read from the curve on p. 98. Require reading lines and units.
The minimum on the graph must be identified by the y-coordinate of the lowest point, then linked to its x-coordinate; a graphical reading remains approximate.

Independent work

Redo practice exercise 2 with the following data: replace the initial value with 265 and the rate or common ratio with 7. Write four lines: data, formula, calculation, conclusion.

Detailed solutions · teacher edition

The solution depends on the exercise revisited. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Method

Lesson plan for 1 h: 7 min of fluency; 10 min of sharing and written summary; 28 min of core exercises; 5 min of correction; 3 min for the exit ticket.

Lesson plan for 2 h: add 20 min of activity work, 20 min of formalising the lesson, a second independent exercise and 10 min of comparative correction.

Differentiated teaching: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast students to produce a counterexample, a justification or a second method.

Session 30 — Review and assessment

Reciprocal function

21/01/2027 • B : 1 h – C : 2 h

Objectives of the session

· Use domain, derivative and sign.

· Write up a study.

· Correct methodically.

Fluency · 6 to 8 minutes, without calculator

1. Calculate 5% of 100. 5

2. Solve $(x-2)(x+1)=0$. $x=-1$ **or** $x=2$

3. Give the forbidden value of $\dfrac{x+1}{x}$. $x=0$

4. Calculate $\dfrac{1}{5}$ and ~~$\dfrac{1}{5}$~~. $\dfrac{1}{5}$ and $-\dfrac{1}{5}$

5. Differentiate 3 x^2-5x+1. $6x-5$

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: use domain, derivative and sign and write a study.

Investigation

Correct a deliberately flawed study of $h(x)=x+\dfrac{4}{x}$: stated domain $\mathbb{R}$, derivative $1+\dfrac{4}{x^2}$, table with no break at 0. Rewrite a complete solution.

Detailed solutions · teacher edition

The domain is $\mathbb{R}\setminus\{0\}$ and $h'(x)=1-\dfrac{4}{x^2}$. The table is split at 0 and takes into account -2 and 2.
Have students verbalise the information that triggers the choice of method.

Written summary

Check the four items separately: domain, expression of the derivative, sign of the derivative, variations. An error at one step must not hide the following ones.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Detailed solutions · teacher edition

The domain is $\mathbb{R} \setminus \{0\}$ and $h'(x) = 1 - \dfrac{4}{x^2}$. The table is split at 0 and takes into account -2 and 2.

Practice · essentials

1. Give the domain and the asymptotes of $f(x) = 2 + \dfrac{4}{x}$.
2. Calculate $f'(x)$ and study its sign.
3. Study on $]0, +\infty[$ the function $g(x) = x + \dfrac{4}{x}$.

Detailed solutions · teacher edition

The domain is $\mathbb{R} \setminus \{0\}$; asymptotes $x = 0$ and $y = 2$. $f'(x) = -\dfrac{4}{x^2} < 0$. For g, $g'(x) = 1 - \dfrac{4}{x^2}$ and the minimum is attained at $x = 2$, with value 4.

References in the textbook

Extensions: Calao, pp. 100-103. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and the digital resources of the textbook.

Calao textbook · p. 98, Exercise 32

Each week, a liquid detergent company estimates that the production cost (in euros) can be modelled by a function C given by: $C(x) = x^2 + 60x + 121$, where x is the volume of detergent (in m ^{3}), $x \in [1;30]$.
Let f the function representing the average cost of production per m ^{3} of liquid detergent.

1. Show that for every $x \in [1;30]$, $f(x) = x + 60 + \dfrac{121}{x}$.
2. Calculate $f'(x)$. Show that for every $x \in [1;30]$, $f'(x) = \dfrac{(x-11)(x+11)}{x^2}$.
3. Draw the variation table of f.
4. What is the minimum average production cost? For what quantity of detergent is it reached ?

Detailed solutions · teacher edition

$f(x) = C(x)/x = x + 60 + 121/x$. Then $f'(x) = 1 - 121/x^2 = ((x-11)(x+11))/x^2$. On $[1;30]$, f' is negative up to 11, then positive. The minimum is attained at $x = 11$ and is $f(11) = 82$ € per m^3.

Independent work

Redo practice exercise 3 with the following data: replace the initial value with 270 and the rate or common ratio with 2. Write four lines: data, formula, calculation, conclusion.

Detailed solutions · teacher edition

The solution depends on the exercise revisited. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Method

Lesson plan for 1 h: 7 min of fluency; 10 min of sharing and written summary; 28 min of core exercises; 5 min of correction; 3 min for the exit ticket.
Lesson plan for 2 h: add 20 min of activity work, 20 min of formalising the lesson, a second independent exercise and 10 min of comparative correction.
Differentiated teaching: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast students to produce a counterexample, a justification or a second method.